Complex Numbers - Loci in the Argand Diagram

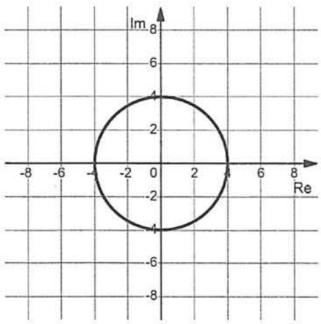
Questions 1–40

Question 1. Sketch the locus of z on a separate Argand diagram in each case, including identifying the maximum and minimum real and imaginary values of z:

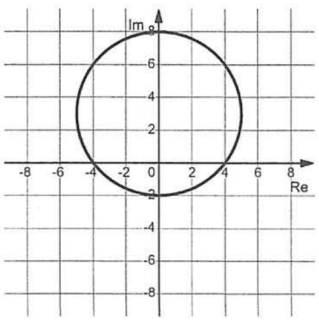
a) $|z| = 6$
b) $|z - 1| = 5$
c) $|z - 2i| = 4$
d) $|z - (4 + 2i)| = 2$
e) $|z - (1 - 3i)| = 5$
f) $|z - 1 + i| = 6$
g) $|z - 4 + 2i| = \sqrt{10}$
h) $|z + 3 - 4i| = \sqrt{5}$

Question 2. For each of the following Argand diagrams, identify the locus of z:

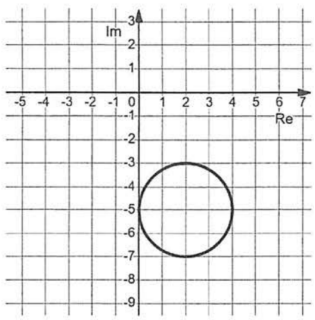
a)



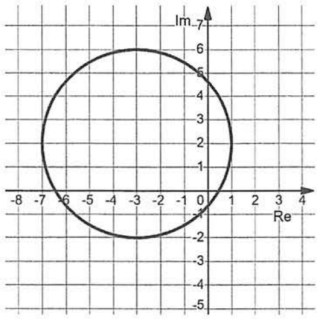
b)



c)



d)



Question 3. For each of the following, find the Cartesian equation of the locus of z and sketch the locus of z, identifying any values where the locus intersects with the axes:

a) $|z+3|=2$
b) $|z+5i|=6$
c) $|z-(-3-i)|=1$
d) $|z-(-1+4i)|=\sqrt{3}$
e) $|z+2-6i|=7$
f) $|1+2i-z|=4$

Question 4. It is known that the complex number z satisfies

$$|z-1-4i|=7$$

a) Sketch the locus of z on an Argand diagram.
b) Find the exact values for which both $|z-1-4i|=7$ and $\mathrm{Re}(z)=0$
c) Find the exact values for which both $|z-1-4i|=7$ and $\mathrm{Im}(z)=0$

Question 5. It is known that the complex number z satisfies

$$|z - 4 - 4i| = 3$$

a) Sketch the locus of z on an Argand diagram.
b) Find the Cartesian equation of this locus.
c) Find the maximum value and minimum value of $\arg(z)$ in the interval $(-\pi, \pi]$ giving your answer to 3 significant figures.
d) Find the maximum and minimum values of $|z|$ for points on the locus.

Question 6. It is known that the complex number z satisfies

$$|z - 3 + 5i| = 2$$

a) Sketch the locus of z on an Argand diagram.
b) Find the Cartesian equation of this locus.
c) Find the maximum value and minimum value of $\arg(z)$ in the interval $(-\pi, \pi]$ giving your answer to 3 significant figures.
d) Find the maximum and minimum values of $|z|$ for points on the locus.

Question 7.

a) Sketch the two loci given by

$$|z-5|=3 \text{ and } |z-1|=5$$

on the same Argand diagram.

b) Find the point of intersection of the two loci in the form $a+bi$ where a and b are real values.

Question 8. Two complex numbers are given by

$$z_1=-2+4i \text{ and } z_2=6-2i$$

It is known that z_1 and z_2 are the ends of a diameter of a circle.

a) Find the equation of the circle in the form

$$|z-a-bi|=r$$

where a, b and r are real values to be found.

There is a complex number w which satisfies

$$|w-p-qi|=4$$

It is known that $6+4i$ is the only complex number which satisfies both loci and that for all points on the locus, $\text{Re}(w)>0$ and $\text{Im}(w)>0$

b) Find the values of p and q

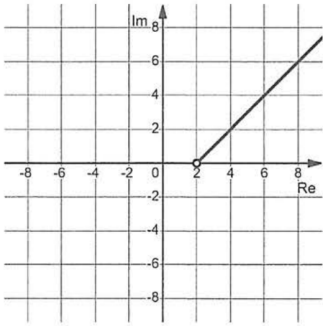
Half-lines

Question 9. Sketch the locus of z on a separate Argand diagram in each case, without identifying any values at which the locus intersects with the real or imaginary axes:

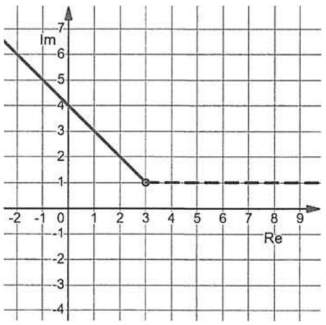
a) $\arg(z) = \frac{\pi}{6}$
b) $\arg(z - 3i) = \frac{\pi}{9}$
c) $\arg(z - 4) = -\frac{\pi}{4}$
d) $\arg(z - (2 + i)) = \frac{3\pi}{4}$
e) $\arg(z - (2 - 4i)) = -\frac{\pi}{3}$
f) $\arg(z - 5 + 3i) = \frac{\pi}{5}$
g) $\arg(z + 1 + 2i) = -\frac{\pi}{5}$
h) $\arg(z + 3 - 2i) = -\frac{3\pi}{5}$

Question 10. For each of the following Argand diagrams, identify the locus of z:

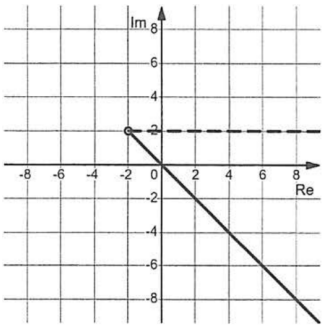
a)



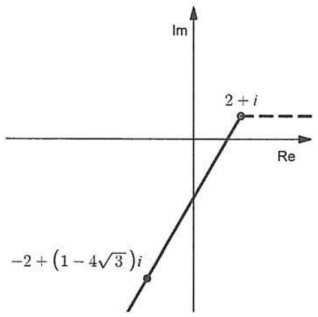
b)



c)



d)



Question 11. Sketch the locus of z, identifying any values where the locus intersects with the real or imaginary axes and find the Cartesian equation of the locus of z for:

a) $\arg(z) = \frac{\pi}{2}$
b) $\arg(z+2) = -\frac{\pi}{4}$
c) $\arg(z+i) = \frac{2\pi}{3}$
d) $\arg(z-(-2-5i)) = -\frac{\pi}{6}$
e) $\arg(z-(-3+3i)) = \frac{7\pi}{8}$
f) $\arg(z+2-4i) = -\frac{\pi}{9}$

Question 12. Given that the complex number z satisfies both

$$\arg(z) = \frac{\pi}{4} \text{ and } \arg(z-3) = \frac{\pi}{2}$$

a) Sketch the locus of $\arg(z) = \frac{\pi}{4}$ and $\arg(z-3) = \frac{\pi}{2}$ on the same diagram.
b) Find z in the form $a+bi$
c) Hence, find $\arg(z-2i)$ giving your answer to 3 significant figures.

Question 13. It is known that the complex number z satisfies

$$\arg(z + 2i) = \frac{\pi}{3}$$

a) Sketch the locus of z on an Argand diagram.

The complex number w lies on the locus of z given above such that $|z|$ is a minimum.

b) Find w in the form $a + bi$ where a and b are real values.

c) Give the minimum value of $|z|$

Question 14. The complex number z satisfies $\arg(z + 6i) = \frac{\pi}{4}$ Additionally, it is given that $w = 1 + 2i$

a) Sketch the locus of z and the point w on an Argand Diagram.

b) Find the minimum distance between w and the locus of z giving your answer in exact form.

Question 15. The complex number z satisfies $\arg(z-2+2i)=\frac{2\pi}{3}$. Given also $w=2\sqrt{3}+2i$

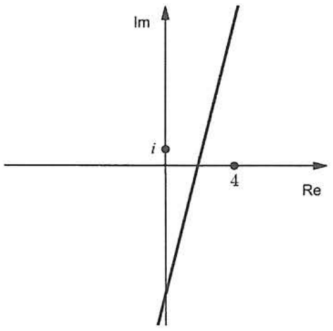
a) sketch the locus of z and the point w on an Argand Diagram.

b) Find the minimum distance between w and the locus of z giving your answer in the form $a+b\sqrt{3}$ where a and b are integers to be found.

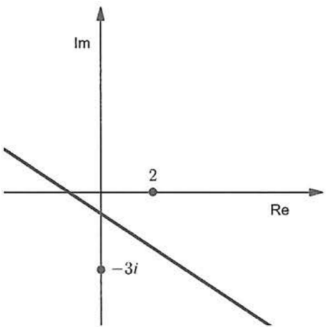
Lines

Question 16. For each of the following Argand diagrams, identify the locus of z using the identified points, and find the cartesian equation of the locus of z in the form $y=mx+c$:

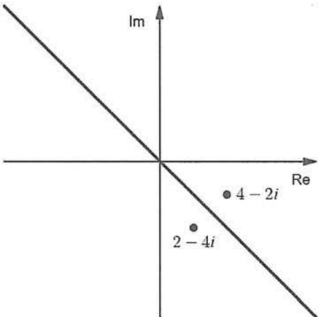
a)



b)



c)



Question 17. Sketch the locus of z, identifying any values where the locus intersects with the real or imaginary axes and find the Cartesian equation of the locus of z in the form $y = mx + c$, for each of the following:

a) $|z - 2| = |z - 4|$
b) $|z + 3 + 5i| = |z + 3 - 2i|$
c) $|z - 2| = |z - i|$
d) $|z - 2| = |z + 3i|$
e) $|z - (-5 - i)| = |z + (2 - 3i)|$
f) $|z + 5 - 3i| = |z - 4 + 7i|$

Question 18.

a) Sketch the two loci given by

$$|z - 2 + i| = |z - 3 + 2i|$$

and

$$|z - 1| = |z + 4i|$$

on the same Argand diagram.

b) Find the value of the complex number z which satisfies both of the loci above, in the form $a + bi$ where a and b are real values.

Question 19. It is known that the complex number z satisfies

$$|z+4| = |z-7i|$$

a) Sketch the locus of z on an Argand diagram. Label any intersections with the real and imaginary axes.
b) Find the Cartesian equation of this locus.
c) Find the least possible value of $|z|$ giving your answer in exact form.

Question 20. It is known that the complex number z satisfies

$$|z-2+3i| = |z-4+4i|$$

a) Sketch the locus of z on an Argand diagram. Label any intersections with the real and imaginary axes.

Furthermore, the complex number z satisfies

$$\arg(z) = -\frac{\pi}{4}$$

b) Find z in the form $a+bi$

Question 21.

a) Show on Argand diagram the solutions to

$$|z+1| = |z-6| = |z-2i|$$

b) Hence find the complex number z which satisfies $|z+1| = |z-6| = |z-2i|$

Mix

Question 22. In each case, sketch on an Argand diagram the locus of z given both conditions, label the possible value(s) of z:

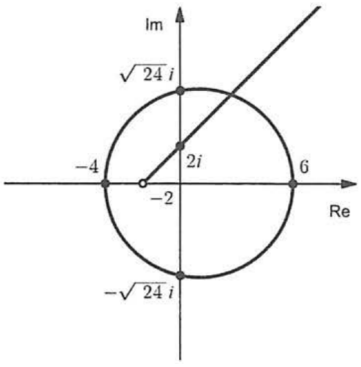
a) $|z| = 3$ and $\arg(z+2) = \frac{\pi}{4}$
b) $|z-3-3i| = 3$ and $\arg(z-3-3i) = \frac{\pi}{3}$
c) $|z-3i| = |z-5i|$ and $\arg(z-1-i) = \frac{3\pi}{4}$
d) $|z+5+2i| = 4$ and $\arg(z-1) = -\frac{3\pi}{4}$
e) $|z+1| = |z+5|$ and $\arg(z+2i) = -\frac{2\pi}{3}$
f) $\arg(z-2+i) = \frac{\pi}{6}$ and $|z-3i| = 2$
g) $|z-2+2i| = 1$ and $|z-2| = |z-2+4i|$
h) $|z+1| = 4$ and $|z+1-i| = \left|z+2+\frac{1}{2}i\right|$
i) $\arg(z-2) = \frac{\pi}{4}$, $|z-1| = |z-i|$ and $|z-2-2i| = 5$

Question 23. Show on Argand diagram the solutions to

$$|z-3| = |z+1| = 4$$

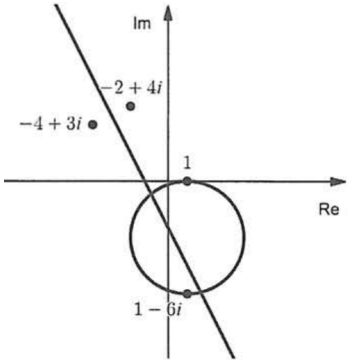
Label the two solutions P and Q

Question 24. For the diagram below:



a) Find the complex locus such that a complex number z_1 lies on the circle in the diagram above.
b) Find the complex locus such that a complex number z_2 lies on the half-line in the diagram above.
c) Find the complex number z such that z satisfies both complex loci above, giving your answers in the form $a+bi$

Question 25. For the diagram below:



a) Find the complex locus such that a complex number z_1 lies on the circle in the diagram above.
b) Find the complex locus such that a complex number z_2 lies on the line in the diagram above.
c) Find the complex numbers which satisfy both complex loci above, giving your answers in the from $a + bi$

Question 26. By first sketching the two loci on the same Argand diagram, find the value of the complex number z such that

$$\left\{z : |z - i| = 2\sqrt{3}\right\} \cap \left\{z : \arg(z + 2 - i) = \frac{\pi}{3}\right\}$$

Question 27. By first sketching the two loci on the same Argand diagram, find the value of the complex number z such that

$$\{z : |z| = 4\} \cap \left\{z : \arg(z - 4\sqrt{2}) = \frac{3\pi}{4}\right\}$$

Question 28. It is known that the complex number z satisfies

$$|z - 3 - 3i| = 3$$

a) Sketch the locus of z on an Argand diagram.

b) Find $|z|$ given that z satisfies

$$|z - 3 - 3i| = 3$$

and

$$\arg(z - 6) = \frac{3\pi}{4}$$

giving your answer in exact form.

Question 29. It is known that the complex number z satisfies

$$|z-2|=4$$

a) Sketch the locus of z on an Argand diagram, label any points that intersect with the real and imaginary axes.

It is also known that z satisfies

$$\arg(z-2)=-\frac{\pi}{4}$$

b) Express z in the form $a+bi$ where a and b are real numbers.

c) Find $|z|^2$ in its simplest form.

Question 30. Sketch the locus of the complex number z given

$$|z-a-bi|=k$$

where $a=b$, $a>0$ and $k<a$

In each case, label the point(s) z given additionally that:

a) $\arg(z-a-bi)=\frac{\pi}{3}$

b) $\arg(z-a-bi)=-\frac{\pi}{4}$

c) $|z-a|=|z-bi|$

d) $|z|=a$

e) $\arg(z)$ is a maximum

f) $\arg(z)$ is a minimum

Question 31. It is known that the complex number z lies on the locus given by

$$\arg(z - 2i) = \frac{\pi}{4}$$

a) Sketch the locus of z on an Argand diagram.
b) Find an equation of a circle with centre $4 + 3i$ that touches the locus in part a), giving your answer in complex form.

Question 32. The complex number z satisfies

$$|z - 2 - 3i| = |z - 5|$$

and

$$|z - ki| = 2$$

where k is a real number. Find the interval in which k must lie.

Question 33. The complex number z satisfies

$$\arg(z-3-ki)=\frac{3\pi}{4}$$

and

$$|z-1-ki|=|z-2|$$

where k is a positive real number. Find the interval in which k must lie.

Question 34. It is known that the complex number z satisfies

$$|z-2-3i|=k$$

and

$$\arg(z-4)=\frac{3\pi}{4}$$

where k is a real number. Find the exact range of values of k

Question 35. Given that the complex number z satisfies

$$|z - 6 - 5i| = 1$$

a) Sketch the locus of z on an Argand diagram.
b) Show that the maximum value of

$$\theta = \arg(z - 2 - 4i)$$

for $-\pi < \theta \leq \pi$ is $\arctan\left(\frac{8}{15}\right)$

Question 36. Given that the complex number z satisfies

$$|z + 4 - 6i| = 3$$

a) Sketch the locus of z on an Argand diagram.
b) Show that the minimum value of

$$\theta = \arg(z - 3 - 3i)$$

for $-\pi < \theta \leq \pi$ is $\frac{\pi}{2} + \arctan\left(\frac{20}{21}\right)$

Question 37. Given that the complex number z satisfies

$$|z + 3 - 5i| = 2$$

a) Sketch the locus of z on an Argand diagram.
b) Find the range of values of θ for which z also satisfies $\theta = \arg(z - 2 - 3i)$ where $\pi < \theta \leq \pi$ giving your answer to 3 significant figures.

Question 38. Given that the complex number z satisfies

$$|z + 1| = 4 \text{ and } \theta = \arg(z - 1)$$

where $-\pi < \theta \leq \pi$
a) find the values of θ such that z additionally satisfies $|z - 2| = 2$
b) Hence state the range of values of θ such that $|z - 2| < 2$

Question 39. The complex number z satisfies

$$|z| = |z - k^2 + 4i|$$

and

$$\arg(z - 2k + 2i) = \frac{\pi}{3}$$

where k is a real number.

a) Show that

$$\mathrm{Re}(z) = \frac{k^4 - 16\sqrt{3}k}{2k^2 - 8\sqrt{3}}$$

b) Find $\mathrm{Re}(z)$ when $\mathrm{Im}(z) = 0$ giving your answer to 3 significant figures. State the value(s) of k to 3 significant figures for when these complex numbers occur.

c) Explain with reference to the loci why the solution to your part a) is not valid when $k = 0$ and $k = 4$

d) Find the range of values of k such that z exists.

Question 40.

a) For $z = x + iy$, show that

$$\frac{z-4}{z} = \frac{x^2 - 4x + y^2}{x^2 + y^2} + \frac{4y}{x^2 + y^2}i$$

It is known that the complex number z satisfies

$$\arg\left(\frac{z-4}{z}\right) = \frac{\pi}{3}$$

b) Show that the equation $\arg\left(\frac{z-4}{z}\right) = \frac{\pi}{3}$ describes the locus of part of a circle.

c) Sketch the locus of z in the Argand diagram.